Complete Experimental Results

Offshore Production Monitoring Benchmark

April 2026

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1 3W Fault Classification

Dataset: Petrobras 3W v2.0.0 — 208,973 sliding windows (720 timesteps × 27 sensors), compressed to (14 × 27) statistical feature matrices, 10 classes. Outer split: 80/20 temporal holdout (n_{test} = 41,515). Inner CV: 5-fold stratified.

Table 1: 3W fault classification — held-out test and inner 5-fold CV. Best test values per column in **bold**. FM = foundation model (zero-shot). Tree = Random Forest ensemble. †After 30-trial Optuna HPO.

	Held-out Test				**CV Accuracy**
Model	**Acc.**	**F_1m**	**F_1w**	**AUC-PR**	mean ± std
DeepONet	**.968**	**.964**	**.969**	.934	.956±.013
LSTM	.967	.962	.967	.931	.951±.014
Random Forest[Tree]	.966	.964	.966	**.985**	.979±.016
PatchTST†	.965	.962	.965	.931	.955±.018
FKM-AD (feat.)	.967	.961	.967	.929	.960±.019
FKM-AD (raw)	.949	.938	.949	.888	.936±.021
MambaSL (feat.)†	.966	.961	.966	.927	.952±.012
MambaSL (raw)†	.919	.887	.920	.807	.930±.010
TiRex[FM]	.912	.895	.911	.938	.899±.021
Majority baseline	*.174*	*.101*	—	—	—

†30-trial Optuna HPO. All others use manually tuned hyperparameters or defaults.

Table 2: Optuna HPO results on 3W (30 trials per model; 33 for DeepONet; 5-fold inner CV). *Best CV F_1m* is the objective optimized by Optuna. *Retrain* metrics are from a model retrained on the full training pool with best hyperparameters and evaluated on the held-out test set.

Model	**Best CV F_1m**	**Best Hyperparameters**	**Retrain Acc.**	**Retrain F_1m**
PatchTST	.949	d=128, heads=16, layers=4, lr=6.8e−5, drop=.24	.965	.962
DeepONet	.941	rank=64, lr=1.3e−3, drop=.08, width=64	.968	.964
LSTM	.936	hidden=512, layers=3, drop=.23, lr=2.4e−4	.966	.961
FKM-AD	30 trials	d_{model}=256, *rank*=32, *lr*=1.6e−3, *drop*=.39	.967	.960

2 Ganymede Gas Production Forecasting

Dataset: 7 wells on the Norwegian Continental Shelf, 31,359 samples, 63 features. Input windows: 90 timesteps. Horizons: $h \in \{7, 14, 30, 90\}$ days. Outer split: temporal 80/20. Inner CV: 3-fold expanding window. Units: MMSCF (million standard cubic feet).

Table 3: Ganymede forecasting — held-out test (multi-well, MMSCF). Best values per column in **bold**. FM = foundation model (zero-shot).

Model	**Metric**	h**=7**	h**=14**	h**=30**	h**=90**
TimesFM$^{\text{FM}}$	R^2_{prod}	**.826**	.763	.656	.413
	MAE	.740	.868	.949	1.02
TiRex$^{\text{FM}}$	R^2_{prod}	.822	**.775**	**.699**	**.565**
	MAE	**.658**	**.749**	**.804**	**.831**
Chronos$^{\text{FM}}$	R^2_{prod}	.734	.668	.584	.361
	MAE	.739	.835	.886	.995
PatchTST	R^2_{prod}	.574	.463	.177	−.718
	MAE	1.77	1.98	2.13	2.51
LSTM	R^2_{prod}	.327	.177	.213	−.464
	MAE	1.77	1.95	1.83	2.25
TCN	R^2_{prod}	−4.51	.108	.060	−.044
	MAE	3.98	2.37	2.62	2.34
DeepONet	R^2_{prod}	−5.72	−6.49	−26.8	−14.9
	MAE	4.84	4.99	5.81	5.70

Table 4: Ganymede held-out test — full metrics per model per horizon.

		MAE				R^2_{prod}			
Model	**Type**	h=7	h=14	h=30	h=90	h=7	h=14	h=30	h=90
TimesFM	FM	0.740	0.868	0.949	1.021	0.826	0.763	0.656	0.413
TiRex	FM	0.658	0.749	0.804	0.831	0.822	0.775	0.699	0.565
Chronos	FM	0.739	0.835	0.886	0.995	0.734	0.668	0.584	0.361
PatchTST	NN	1.771	1.980	2.134	2.510	0.574	0.463	0.177	−0.718
LSTM	NN	1.768	1.948	1.833	2.254	0.327	0.177	0.213	−0.464
TCN	NN	3.982	2.369	2.623	2.336	−4.509	0.108	0.060	−0.044
DeepONet	NN	4.836	4.987	5.812	5.700	−5.716	−6.495	−26.79	−14.93

Table 5: Ganymede 3-fold expanding window CV (multi-well, mean ± std). Top: R^2_{prod}, bottom: MAE.

Model	**Metric**	***h*=7**	***h*=14**	***h*=30**	***h*=90**
TimesFM$^{\text{FM}}$	R^2_{prod}	$.69 \pm .07$	$.60 \pm .10$	$.48 \pm .17$	$.35 \pm .28$
TiRex$^{\text{FM}}$	R^2_{prod}	$.74 \pm .07$	$.67 \pm .13$	$.60 \pm .17$	$.49 \pm .22$
Chronos$^{\text{FM}}$	R^2_{prod}	$.63 \pm .05$	$.54 \pm .12$	$.43 \pm .19$	$.31 \pm .24$
PatchTST	R^2_{prod}	$.56 \pm .40$	$.39 \pm .60$	$.14 \pm .91$	$-.23 \pm 1.34$
LSTM	R^2_{prod}	$.69 \pm .17$	$.54 \pm .34$	$.41 \pm .47$	$.20 \pm .57$
TCN	R^2_{prod}	-7.96 ± 12.29	-1.07 ± 2.52	-3.52 ± 5.93	-6.68 ± 10.31
DeepONet	R^2_{prod}	$-.03 \pm 1.07$	$-.17 \pm 1.16$	-10.9 ± 16.4	-27.9 ± 40.2
TimesFM$^{\text{FM}}$	MAE	3.54 ± 1.97	3.96 ± 2.19	4.28 ± 2.38	4.68 ± 3.45
TiRex$^{\text{FM}}$	MAE	3.11 ± 1.75	3.34 ± 1.84	3.57 ± 1.97	3.99 ± 2.83
Chronos$^{\text{FM}}$	MAE	3.42 ± 1.93	3.65 ± 2.02	3.93 ± 2.18	4.35 ± 2.97
PatchTST	MAE	4.16 ± 1.41	4.24 ± 1.33	4.56 ± 1.48	4.70 ± 1.88
LSTM	MAE	3.78 ± 1.88	4.08 ± 1.69	4.59 ± 1.96	4.81 ± 2.68
TCN	MAE	5.28 ± 0.72	5.05 ± 0.85	5.15 ± 1.10	5.79 ± 1.61
DeepONet	MAE	5.26 ± 1.56	5.36 ± 2.06	6.74 ± 0.36	7.50 ± 1.86

Table 6: Ganymede held-out test — RMSE and MASE per model per horizon (multi-well).

Model	**Metric**	***h*=7**	***h*=14**	***h*=30**	***h*=90**
TimesFM$^{\text{FM}}$	RMSE	1.389	1.591	1.737	1.888
TiRex$^{\text{FM}}$	RMSE	1.398	1.525	1.602	1.635
Chronos$^{\text{FM}}$	RMSE	1.623	1.764	1.820	1.932
PatchTST	RMSE	2.192	2.441	2.717	3.307
LSTM	RMSE	2.616	2.876	2.546	3.090
TCN	RMSE	7.082	3.201	3.479	3.032
DeepONet	RMSE	7.910	8.196	13.360	9.686
TimesFM$^{\text{FM}}$	MASE	1.681	1.012	1.117	1.415
TiRex$^{\text{FM}}$	MASE	1.496	0.873	0.946	1.152
Chronos$^{\text{FM}}$	MASE	1.679	0.974	1.043	1.379
PatchTST	MASE	4.025	2.308	2.511	3.479
LSTM	MASE	4.018	2.271	2.158	3.123
TCN	MASE	9.050	2.761	3.088	3.236
DeepONet	MASE	10.99	5.814	6.842	7.899

2.1 Per-Well Results

Table 7: Ganymede R^2_{prod} per well — h=7 (held-out test). Best per well in **bold**.

Model	**Z01Z**	**Z02Z**	**Z04**	**Z05Z**	**Z06**	**Z07**	**Z08**
TimesFM$^{\text{FM}}$	−0.788	−0.342	**0.153**	0.001	**0.279**	**0.208**	−0.517
TiRex$^{\text{FM}}$	**−0.749**	**−0.248**	0.129	**0.179**	0.205	0.117	**−0.830**
Chronos$^{\text{FM}}$	−0.832	−0.467	−0.122	−0.050	0.114	−0.154	−1.767
PatchTST	−2266	−0.213	−0.345	−151.5	−7.139	−1.339	−0.825
LSTM	−173.8	−0.563	−5.823	−0.659	−1.337	0.012	−3.831
TCN	−48.04	−2.491	−9.858	−7.514	−6.702	−0.347	−8.158
DeepONet	−110.6	−1.307	−34.59	−112.8	−60.39	−0.447	−100.2

Table 8: Ganymede R^2_{prod} per well — h=14 (held-out test). Best per well in **bold**. TCN per-well results not available for h=14.

Model	**Z01Z**	**Z02Z**	**Z04**	**Z05Z**	**Z06**	**Z07**	**Z08**
TimesFM$^{\text{FM}}$	**−1.101**	−0.822	**−0.022**	−0.579	−0.702	−0.050	**−0.423**
TiRex$^{\text{FM}}$	−1.030	**−0.714**	−0.075	**−0.213**	−0.771	−0.155	−0.722
Chronos$^{\text{FM}}$	−1.162	−0.877	−0.360	−0.469	**−0.885**	−0.498	−1.582
PatchTST	−3963	−0.579	−0.478	−187.1	−57.43	−1.964	−1.288
LSTM	−6.691	−0.394	−1.007	−1.107	−5.142	**0.300**	−0.996
DeepONet	−500.6	−4.950	−11.62	−87.01	−346.9	−1.203	−88.77

3 CDF Anomaly Detection

Dataset: Cumulative Distribution Function reconstruction task. Outer split: temporal holdout ($n_{\text{test}} = 864$). Inner CV: 3-fold. Metric: reconstruction error (lower is better).

Table 9: Ganymede R^2_{prod} per well — h=30 (held-out test). Best per well in **bold**. TCN per-well results not available for h=30.

Model	**Z01Z**	**Z02Z**	**Z04**	**Z05Z**	**Z06**	**Z07**	**Z08**
TimesFM$^{\text{FM}}$	**−1.149**	**−1.288**	**−0.356**	−1.323	**−0.595**	**−0.260**	**−1.212**
TiRex$^{\text{FM}}$	−1.104	−1.188	−0.477	**−0.693**	−0.686	−0.408	−1.357
Chronos$^{\text{FM}}$	−1.213	−1.398	−0.898	−0.834	−0.855	−0.764	−1.490
PatchTST	−4761	−0.826	−1.013	−244.1	−111.1	−3.556	−1.795
LSTM	−9.739	−1.154	−0.549	−4.927	−3.401	−0.250	−17.17
DeepONet	−487.3	−8.848	−19.63	−55.93	−101.2	−1.308	−710.3

Table 10: Ganymede R^2_{prod} per well — h=90 (held-out test). Best per well in **bold**. TCN per-well results not available for h=90.

Model	**Z01Z**	**Z02Z**	**Z04**	**Z05Z**	**Z06**	**Z07**	**Z08**
TimesFM$^{\text{FM}}$	**−1.059**	−1.629	**−0.693**	−2.690	**0.088**	−0.451	−1.385
TiRex$^{\text{FM}}$	−1.064	**−1.640**	−0.716	**−2.020**	−0.032	−0.463	**−1.384**
Chronos$^{\text{FM}}$	−1.115	−1.548	−1.450	−2.211	−8.270	−0.987	−0.885
PatchTST	−4372	−0.983	−1.137	−267.5	−991.2	−5.227	−2.315
LSTM	−58.83	−1.384	−1.801	−0.919	−66.27	−0.372	−1.422
DeepONet	−202.1	−6.284	−17.52	−26.02	−694.3	**−0.603**	−1.861

4 Cross-Dataset Summary

Table 11: Ganymede MAE per well — h=7 (held-out test, MMSCF). Best per well in **bold**.

Model	**Z01Z**	**Z02Z**	**Z04**	**Z05Z**	**Z06**	**Z07**	**Z08**
TimesFM$^{\mathrm{FM}}$	0.025	1.846	0.531	0.264	0.353	1.485	0.707
TiRex$^{\mathrm{FM}}$	**0.024**	**1.647**	**0.483**	**0.236**	**0.295**	**1.381**	**0.738**
Chronos$^{\mathrm{FM}}$	**0.024**	1.701	0.516	0.253	0.299	1.457	0.838
PatchTST	4.435	2.201	0.878	4.336	4.311	2.302	1.786
LSTM	0.982	2.309	1.521	0.390	1.695	1.974	2.835
TCN	0.524	3.394	1.970	0.824	3.921	2.175	3.554
DeepONet	0.787	3.305	2.893	2.848	7.069	2.176	9.681

Table 12: CDF anomaly detection — held-out test reconstruction error. Lower is better. Best values in **bold**. FM = foundation model (zero-shot).

	Held-out Test					**CV**
Model	**Mean**	**Std**	**P50**	**P95**	**P99**	**Mean ± Std**
LSTM	**0.005**	**0.003**	**0.004**	**0.009**	**0.014**	0.099 ± 0.104
PatchTST	0.061	0.061	0.047	0.146	0.235	0.262 ± 0.137
DeepONet	0.219	0.123	0.186	0.449	0.503	0.975 ± 0.605
Chronos$^{\mathrm{FM}}$	24.81	19.19	19.66	60.92	82.89	24.83 ± 3.86
TimesFM$^{\mathrm{FM}}$	24.88	21.00	19.07	64.89	90.24	23.65 ± 3.65
TiRex$^{\mathrm{FM}}$	24.88	20.48	19.21	63.19	85.22	21.93 ± 3.22
Mean baseline	*41.46*	*17.51*	*39.68*	*66.26*	*93.90*	—

Table 13: Model performance summary across all three tasks. Arrows indicate metric direction (↑ = higher is better, ↓ = lower is better). Best in **bold**. N/A = model not evaluated on task.

	3W (Classification)		**Ganymede (h=7)**		**CDF**
Model	Acc. ↑	F_1m ↑	R^2_{prod} ↑	MAE ↓	Error ↓
DeepONet	**.968**	**.964**	−5.72	4.84	0.219
LSTM	.967	.962	.327	1.77	**0.005**
Random Forest$^{\mathrm{Tree}}$	.966	.964	N/A	N/A	N/A
PatchTST	.965	.962	.574	1.77	0.061
FKM-AD (feat.)	.967	.961	N/A	N/A	N/A
FKM-AD (raw)	.949	.938	N/A	N/A	N/A
MambaSL (feat.)†	.966	.961	N/A	N/A	N/A
MambaSL (raw)†	.919	.887	N/A	N/A	N/A
TCN	N/A	N/A	−4.51	3.98	N/A
TiRex$^{\mathrm{FM}}$	.912	.895	**.822**	**.658**	24.88
TimesFM$^{\mathrm{FM}}$	N/A	N/A	.826	.740	24.88
Chronos$^{\mathrm{FM}}$	N/A	N/A	.734	.739	24.81

Table 14: Key findings per dataset.

Dataset	**Winner**	**Key Finding**
3W	Trained models & Random Forest	DeepONet 96.8%, LSTM 96.7%, FKM-AD 96.7%, RF 96.6%, MambaSL 96.6%, PatchTST 9 RF, FKM-AD & MambaSL prove feature extraction captures structure
Ganymede	Foundation models	TiRex/TimesFM $R^2_{\mathrm{prod}} > 0.8$ at h=7 TCN $R^2_{\mathrm{prod}} \leq 0.11$ → local receptive fields insufficient
CDF	Trained models (task-specific)	LSTM error 0.005 vs FMs ∼25 FMs lack CDF reconstruction inductive bias